

GRENCHE, Switzerland

WMO: 066320

Lat: 47.1792N

Lon: 7.4150E

Elev: 429

StdP: 96.28

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.6	-6.2	-13.2	1.3	-7.7	-8.7	1.9	-5.1	11.8	2.8	10.5	1.7	4.4	60	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	31.2	20.4	29.3	19.9	27.4	19.1	21.2	28.9	20.5	27.6	19.7	26.0	3.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.5	23.3	18.2	13.8	22.3	17.5	13.2	21.7	63.6	29.0	60.6	27.9	58.0	26.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.2	6.9	DB	-12.0	34.0	3.0	1.7	-14.1	35.3	-15.9	36.3	-17.5	37.3	-19.7	38.5	(4)
(5)			WB	-12.2	23.3	3.1	1.2	-14.5	24.1	-16.2	24.8	-18.0	25.5	-20.2	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	1.4	1.5	6.0	10.1	13.5	18.1	19.9	19.5	15.5	10.8	5.5	1.8	(6)
(7)		DBStd	7.52	3.54	4.06	3.18	3.49	3.45	3.21	3.02	3.10	2.90	3.27	3.10	3.51	(7)
(8)		HDD10.0	1104	268	238	127	41	8	0	0	0	0	31	138	255	(8)
(9)		HDD18.3	3088	526	471	382	246	153	43	18	24	94	234	385	513	(9)
(10)		CDD10.0	1236	0	1	3	45	117	243	307	296	165	56	3	0	(10)
(11)		CDD18.3	177	0	0	0	0	3	36	67	61	9	0	0	0	(11)
(12)		CDH23.3	1809	0	0	0	9	65	398	674	574	89	0	0	0	(12)
(13)		CDH26.7	554	0	0	0	1	7	122	229	187	9	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.7	2.7	2.8	2.5	2.3	2.2	1.8	2.0	1.9	2.1	2.4	(14)	
(15)	Precipitation	PrecAvg	1204	86	83	80	92	110	113	120	114	100	99	107	(15)	
(16)		PrecMax	1528	245	209	301	192	202	245	250	263	178	187	232	236	(16)
(17)		PrecMin	846	15	23	22	13	35	51	45	10	44	14	10	0	(17)
(18)		PrecStd	198	56	47	63	50	48	47	44	53	42	45	58	60	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	14.9	19.4	24.7	27.8	32.4	34.5	32.6	27.8	22.1	16.2	11.0	(19)
(20)			MCWB	9.0	8.9	10.9	15.7	17.9	22.3	20.9	21.0	18.5	15.1	12.3	9.1	(20)
(21)		2%	DB	9.2	11.3	17.6	22.1	25.1	30.1	31.3	30.7	26.2	20.0	13.7	9.7	(21)
(22)			MCWB	7.8	7.4	10.3	13.7	16.4	20.7	20.0	20.6	18.3	14.7	11.4	8.5	(22)
(23)		5%	DB	7.5	9.0	15.3	20.2	23.2	27.6	29.3	28.9	23.8	17.8	11.9	8.4	(23)
(24)			MCWB	6.2	6.5	9.3	13.1	15.6	19.4	19.6	19.9	17.2	14.1	10.5	7.3	(24)
(25)		10%	DB	6.2	7.0	12.7	17.8	21.0	25.5	27.1	26.7	21.7	16.0	9.9	6.8	(25)
(26)			MCWB	5.1	5.0	8.3	11.7	14.7	18.3	18.7	19.3	16.4	13.5	8.9	5.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.8	11.7	15.4	18.8	23.3	21.8	22.0	19.8	16.6	13.0	9.7	(27)
(28)			MCDB	10.4	12.7	17.8	23.1	25.6	31.1	30.8	29.8	25.1	19.7	15.0	10.3	(28)
(29)		2%	WB	8.0	7.9	10.6	14.2	17.3	21.4	20.8	21.1	18.9	15.6	11.7	8.7	(29)
(30)			MCDB	9.1	11.2	16.0	21.0	23.4	28.2	29.1	28.7	24.5	18.2	13.3	9.5	(30)
(31)		5%	WB	6.3	6.5	9.6	13.2	16.2	20.1	20.1	20.4	17.9	14.7	10.4	7.4	(31)
(32)			MCDB	7.3	8.8	14.5	19.4	21.5	26.4	27.5	27.5	22.5	17.0	11.7	8.3	(32)
(33)		10%	WB	5.3	5.1	8.6	12.1	15.2	18.9	19.3	19.6	17.0	13.8	9.0	6.0	(33)
(34)			MCDB	6.1	6.7	12.3	17.0	20.0	24.3	25.6	25.6	20.8	15.8	9.8	6.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	7.2	10.5	11.3	10.9	11.2	11.7	10.8	8.5	5.6	4.9	(35)	
(36)			MCDBR	7.2	12.1	16.3	16.3	15.3	15.5	16.0	15.0	14.2	12.1	9.0	6.8	(36)
(37)			MCWBR	5.6	8.3	9.4	8.5	7.3	6.9	5.7	5.7	6.9	7.3	6.6	5.7	(37)
(38)		5% WB	MCDBR	6.9	11.3	14.4	15.1	13.8	14.6	14.2	14.1	12.2	10.4	8.0	6.3	(38)
(39)			MCWBR	5.7	7.9	8.4	8.0	6.9	6.7	5.5	5.6	6.3	6.7	6.2	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.326	0.362	0.400	0.410	0.413	0.413	0.404	0.382	0.358	0.323	0.297	(40)	
(41)		taud	2.440	2.366	2.312	2.231	2.249	2.286	2.280	2.316	2.360	2.409	2.455	2.447	(41)	
(42)		Ebn at Noon	791	837	855	852	855	853	847	839	824	787	752	748	(42)	
(43)		Edh at Noon	67	89	109	130	132	128	127	118	103	85	66	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.17	2.00	3.21	4.35	5.03	5.95	5.74	4.99	3.82	2.42	1.31	0.95	(44)	
(45)		RadStd	0.15	0.31	0.48	0.64	0.48	0.55	0.51	0.47	0.35	0.23	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (111 neighbors)		N/A	N/A	N/A	+0.75	+0.40	+0.30	N/A	N/A	+57	+17

Nomenclature: See separate page